

Chapter 5

FREE CONVECTION

Free convection or natural convection pertains to the motion of a fluid caused by density differences during heating or cooling of the fluid. A common demonstration is the heating of a solution in a beaker in the laboratory. The liquid at the bottom of the beaker gets heated up and its density decreases. The liquid above is at a lower temperature and, therefore, has a higher density. Thus a buoyancy force comes into play; the lighter liquid at the bottom moves up, and the heavier liquid at the top flows down creating free convection currents. The up flowing hot liquid transfers heat from the bottom region to the top region. The phenomenon is called *free convection* or *natural convection*, because no mechanical device or externally imposed force is used to create motion in the fluid.

There are numerous examples of free convection heat transfer in industrial operations. Process equipment and accessories at an elevated temperature (or at a sub-atmospheric temperature) are insulated to avoid heat loss (or gain). The *skin temperature* of the insulation will be more than the ambient temperature. This causes free convection currents on the surface of the insulation and consequent heat loss. Some of the examples are heat losses from steam pipes, reactors vessels, furnaces, etc.

5.1. QUALITATIVE DESCRIPTION OF FREE CONVECTION FLOWS

The buoyancy force that causes convective flow is often called *body force* and is gravitational in nature. This force, i.e. buoyant force comes into play when the fluid is heated at the bottom (or cooled at the top). Such a system is *hydrodynamically unstable* for all practical purposes. When the fluid is heated at the top, for example, by confining the liquid in the region between two horizontal parallel surfaces, the system is *stable*. No convection current sets in because the buoyancy force simply does not exist. This situation is shown in Fig. 5.1. Free convection flow at or over a heated surface—for example, a wall, a wire or a cylinder—occurs in the form of a *plume*. The plume width varies with position as shown in Fig. 5.2.

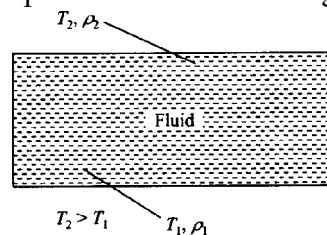


Fig. 5.1 A fluid confined between two parallel surfaces, the upper surface is at a higher temperature.

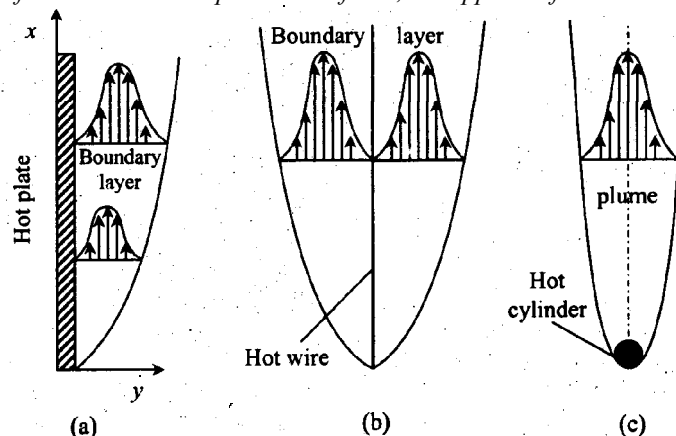


Fig. 5.2 Free convection boundary layer: (a) along a heated plate, (b) along a hot vertical wire, and (c) plume over a hot horizontal cylinder.

However, when a fluid is enclosed between two horizontal surfaces (at a distance L apart, say), the lower surface being hot and the upper surface cold, heat transfer occurs practically by pure conduction when the Grashof number Gr_L , (this is a dimensionless group characterizing free convection; see Section 5.2) is less than about 1700. Above this value of the Grashof number, free convection begins and a pattern of hexagonal prismatic cells is formed as shown in Fig. 5.3. At the middle of each prism, the warmer fluid ascends; the colder fluid descends along the 'walls' of the hexagonal prisms. The top surface looks like the cross-section of a honeycomb. This phenomenon was first observed by Benard in 1901 while heating a thin layer of spermaceti from below. These free convection cells are called *Benard cells*.

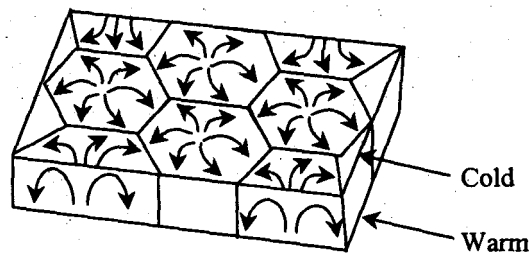


Fig. 5.3 Benard cells in a layer of liquid heated from below (schematic).

Other common situations of free convection heat transfer are from a horizontal (or inclined) plate, hot or cold, facing upwards or downwards. Free convection in a vertical or inclined enclosure made up of two parallel, cylindrical or spherical surfaces is also important from the practical angle. The natural convection flow patterns in a few common geometries are qualitatively shown in Fig. 5.4.

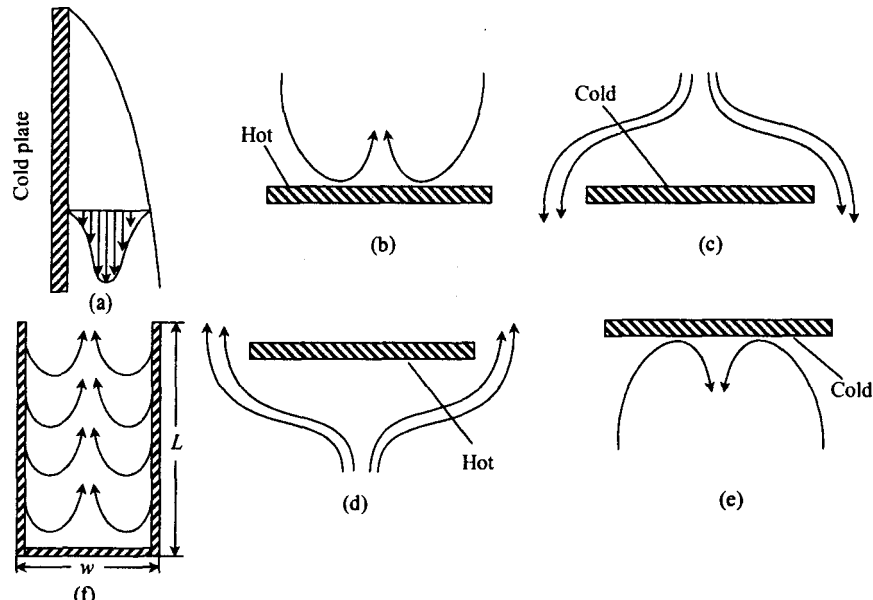


Fig. 5.4 Free convection flow patterns: (a) on a cold vertical surface, (b) on a hot surface facing up, (c) on a cold surface facing up, (d) on a hot surface facing down, (e) on a cold surface facing down, and (f) on a vertical enclosure with cold parallel surfaces.

5.2. HEAT TRANSFER CORRELATIONS FOR FREE CONVECTION

Analytical solutions for velocity and temperature distributions as well as for the heat transfer rate in free convective motion are available for a few simple situations. These solutions are based on the similarity principle (see, for example, Schlichting, 1968) and boundary layer approximations. However, in many practical cases we have to depend on empirical correlations. Free convection

heat transfer correlations involve the Nusselt number, the Prandtl number and the Grashof number. The expression for the Grashof number Gr_L is given below (the details of how the Grashof number arises in the analysis of free convection heat transfer are not given here).

$$Gr_L = \frac{g\beta(T_s - T_o)L^3}{\nu^2} \quad (5.1)$$

where

$$\beta = -\frac{1}{\rho} \left(\frac{\partial \rho}{\partial T} \right)_p = \text{the coefficient of volumetric expansion}$$

g is the acceleration due to gravity

L is a characteristic length

T_o is the ambient fluid temperature

T_s is the surface temperature

ρ is the fluid density

P is the total pressure.

The physical significance of the Grashof number is given below. The Grashof number plays the same role in natural convection as the Reynolds number does in forced convection.

$$\text{Grashof number} = \frac{\text{Bouyancy force}}{\text{Viscous force}}$$

The larger the temperature difference between the wall and the fluid, the greater will be the induced flow. When the fluid velocity is large enough, the free convection boundary layer changes from laminar to turbulent. This transition has a strong influence on the rate of heat transfer. The condition under which the transition takes place is given by a critical value of a dimensionless number called the *Rayleigh number*, Ra_L defined as (where L is the characteristic length)

$$Ra_L = (Gr_L) (Pr) = \frac{g\beta(T_s - T_o)L^3}{\nu\alpha} \quad (5.2)$$

For the case of a vertical surface the *critical Rayleigh number* is taken as $Ra_L = 10^9$. Above this value of the Rayleigh number the flow changes from laminar to turbulent. However, the flow may remain laminar over a part of the surface, and turbulent over the rest (compare with the case of boundary layer flow over a flat plate). The distance from the bottom edge (if the surface is hot) where transition occurs is given by the local Rayleigh number, $Ra_x = 10^9$. A few important correlations for free convection heat transfer from surfaces of common geometries are discussed below. These correlations are usually of the form $Nu = f(Ra, Pr)$, and the relevant physical properties are taken at the mean film temperature.

5.2.1. Free Convection from a Flat Surface

The rate of free convection heat transfer from a flat surface or plate greatly depends upon the orientation of the plate-whether it is vertical, inclined, or horizontal. Again, a horizontal plate may be hot or cold, and the hot or the cold surface may face upwards or downwards. A single correlation does not cover all these situations.

(i) For a vertical plate, hot or cold, the following equation suggested by Churchill and Chu (1975) may be used for the entire range of Rayleigh number.

$$\overline{\text{Nu}}_L = \frac{\bar{h}L}{k} = \left[0.825 + \frac{0.387 (\text{Ra}_L)^{1/6}}{\left\{ 1 + (0.492/\text{Pr})^{9/16} \right\}^{8/27}} \right]^2 \quad (5.3)$$

However, if the free convective flow over the plate is laminar (i.e. $\text{Ra}_L < 10^9$), the following equation gives a better prediction.

$$\overline{\text{Nu}}_L = \frac{\bar{h}L}{k} = 0.68 + \frac{0.67 (\text{Ra}_L)^{1/4}}{\left\{ 1 + (0.492/\text{Pr})^{9/16} \right\}^{4/9}} \quad (5.4)$$

These equations are also applicable for an inclined plate if the angle of inclination with the vertical, $\theta < 60^\circ$. In this chapter, in all the correlations, h denotes the average heat transfer coefficient and Nu the average Nusselt number.

Free convection heat transfer from a flat surface is of much practical importance because heat transfer from furnace walls and many other equipment occurs by this mechanism.

(ii) Different flow situations arise when heat transfer by natural convection occurs from a horizontal plate depending upon whether the plate is hot or cold, or whether the heat transfer surface faces up or down. The nature of flow in such cases has been shown qualitatively before. A characteristic length in all these cases is defined as below.

$$L = \frac{\text{Area of the plate}}{\text{Perimeter of the plate}}$$

A few useful correlations are given below:

(a) Heat transfer *from the upper surface* of a flat plate:

$$\overline{\text{Nu}} = \frac{\bar{h}L}{k} = 0.54 (\text{Ra}_L)^{1/4}; \quad 10^4 \leq \text{Ra}_L \leq 10^7 \quad (5.5)$$

(b) Heat transfer *to the lower surface* of a cold plate:

$$\overline{\text{Nu}} = \frac{\bar{h}L}{k} = 0.15 (\text{Ra}_L)^{1/3}; \quad 10^7 \leq \text{Ra}_L \leq 10^{11} \quad (5.6)$$

(c) Heat transfer *from the lower surface* of a hot plate or *to the upper surface* of a cold plate:

$$\overline{\text{Nu}} = \frac{\bar{h}L}{k} = 0.27 (\text{Ra}_L)^{1/4}; \quad 10^5 \leq \text{Ra}_L \leq 10^{10} \quad (5.7)$$

5.2.2. Free convection from a cylinder

(a) For heat transfer from or to a horizontal cylinder, Morgan (1975) recommended a correlation of the following form

$$\overline{\text{Nu}} = \frac{\bar{h}d}{k} = C(\text{Ra}_d)^n \quad (5.8)$$

where d is the diameter of the cylinder. The constant C and the exponent n may be obtained from Table 5.1 for different ranges of the Rayleigh number.

Table 5.1 Values of C and n for Eq. (5.8)

Ra_d	C	n
$10^{-10} - 10^{-2}$	0.675	0.058
$10^{-2} - 10^2$	1.02	0.148
$10^2 - 10^4$	0.85	0.188
$10^4 - 10^7$	0.48	0.250
$10^7 - 10^{12}$	0.125	0.333

The following correlation by Churchill and Chu (1975) may also be used for free convection heat transfer calculation from a horizontal cylinder.

$$\overline{Nu} = \frac{\bar{h}d}{k} = \left[0.60 + \frac{0.387(Ra_d)^{1/6}}{\left\{ 1 + (0.559/Pr)^{9/16} \right\}^{8/27}} \right]^2; \quad 10^{-5} < Ra_d < 10^{12} \quad (5.9)$$

(b) In the case of a *vertical cylinder*, the equations for a vertical flat plate are applicable if the thickness of the free convection boundary layer is much smaller than the cylinder diameter. The cylinder may be considered to be equivalent to a vertical flat plate of breadth equal to the circumference of the cylinder. The criterion for this is quantitatively given by

$$\frac{d}{L} \geq \frac{35}{(Gr_L)^{1/4}} \quad (5.10)$$

Here L is the height of the vertical cylinder. However, if the cylinder diameter is small and the above condition is not fulfilled, the numerical results of Minkowycz and Sparrow (1974) may be used.

5.2.3. Free Convection from a Sphere

The following correlation suggested by Churchill (1983) may be used.

$$\overline{Nu} = \frac{\bar{h}d}{k} = 2 + \frac{0.589 (Ra_d)^{1/4}}{\left\{ 1 + (0.469/Pr)^{9/16} \right\}^{4/9}}; \quad Pr \geq 0.7; \quad Ra_d \leq 10^{11} \quad (5.11)$$

5.3. COMBINED FREE AND FORCED CONVECTION

There are practical situations in which the contributions of both free and forced convection should be taken into account for heat transfer calculations. The following conditions determine the regimes of free convection, forced convection, and combined free and forced convection (or mixed convection).

$$\frac{Gr_L}{Re_L^2} \ll 1, \text{ forced convection regime (negligible free convection)}$$

$$\frac{Gr_L}{Re_L^2} \gg 1, \text{ free convection regime (negligible forced convection)}$$

$$\frac{Gr_L}{Re_L^2} \sim 1, \text{ mixed convection regime (both free and forced convection are important)}$$

If heat transfer occurs in the mixed convection regime, the following equation may be used to calculate the Nusselt number.

$$\text{Nu}^m = (\text{Nu}_{\text{forced}})^m \pm (\text{Nu}_{\text{free}})^m \quad (5.12)$$

where $\text{Nu}_{\text{forced}}$ and Nu_{free} are Nusselt numbers for forced and free convection, respectively. A value of $m = 3$ is usually recommended. A positive or a negative sign is taken depending upon whether the free convection flow occurs in the same or the opposite direction of forced convection flow.

PROBLEMS

1. A furnace with a steel door, having an inner lining of an insulating material, is at a temperature of 65°C. The door, 1.5 m high and 1 m wide, loses heat to an ambient at 25°C. Calculate the rate of heat loss from the door at steady state.

The average air-film temperature, $T_f = (65 + 25)/2 = 45^\circ\text{C} = 318 \text{ K}$

The relevant properties of the air at this temperature are: Prandtl number, $\text{Pr} = 0.695$; $\nu = 1.85 \times 10^{-5} \text{ m}^2/\text{s}$; $\beta = 1/T_f = 1/318 = 3.145 \times 10^{-3} \text{ K}^{-1}$; $k = 0.028 \text{ W/m } ^\circ\text{C}$.

2. A thin metal plate 1 m $\times$ 1 m is placed on a rooftop. It receives radiant heat from the sun directly at the rate of 170 W/m². If heat transfer from the plate to the ambient occurs purely by free convection, calculate the steady state temperature of the plate. Assume that there is no heat loss from the bottom of the plate. The ambient temperature is 25°C.

Let us assume a plate temperature, $T_s = 60^\circ\text{C}$.

The mean film temperature = $(60 + 25)/2 = 42.5^\circ\text{C} = 315.5 \text{ K}$

Given: $T_s - T_o = 60 - 25 = 35^\circ\text{C}$; Properties of the air at the mean film temperature: $\beta = 1/315.5 = 3.17 \times 10^{-3} \text{ K}^{-1}$; $\nu = 1.85 \times 10^{-5} \text{ m}^2/\text{s}$; $\text{Pr} = 0.7$; $k = 0.028 \text{ W/m } ^\circ\text{C}$, and $g = 9.8 \text{ m/s}^2$.

3. A horizontal steam pipe, 78 mm i.d. and 89 mm o.d., carries saturated steam at 15 kg/cm² gauge pressure. It is lagged only with a 2 cm thick layer of pre-formed mineral fibre of thermal conductivity 0.05 W/m °C which is exposed to ambient air at 25°C. Calculate the rate of heat loss by free convection per metre length of the pipe.

Now we calculate the convection film coefficient. The properties of the air taken at the mean film temperature, i.e. $T_f = (50 + 25)/2 = 37.5^\circ\text{C} (= 310.5 \text{ K})$, are: $\nu = 1.76 \times 10^{-5} \text{ m}^2/\text{s}$; $\beta = (1/310.5) = 3.22 \times 10^{-3} \text{ K}^{-1}$; $k = 0.027 \text{ W/m } ^\circ\text{C}$; $\text{Pr} = 0.705$.

4. A horizontal section of an uninsulated pipe, 60 mm o.d., carrying warm water runs below an exhaust fan in a factory shed. The fan creates a suction that causes a mild upward cross-flow velocity of 0.3 m/s on the pipe. If the pipe wall temperature is 60°C, calculate the rate of heat loss by the combined free and forced convection per metre length of the pipe. The ambient air temperature is 30°C.

For air at 318 K, we have

$\rho = 1.105 \text{ kg/m}^3$; $c_p = 0.24 \text{ kcal/kg } ^\circ\text{C}$; $\mu = 1.95 \times 10^{-5} \text{ kg/m s}$; $\text{Pr} = 0.7$;
 $\nu = \mu/\rho = 1.85 \times 10^{-5} \text{ m}^2/\text{s}$; $k = 0.0241 \text{ kcal/h m } ^\circ\text{C}$; $\beta = 1/318 = 3.145 \times 10^{-3} \text{ K}^{-1}$; and
 cross flow air velocity, $V = 0.3 \text{ m/s}$.